



SAMPLE COMPONENTS

The physics at the heart of the beamline

Diffraction, small-angle scattering, spectroscopy & imaging in McStas & McXtrace

McStas — neutrons

McXtrace — X-rays



Peter Willendrup · DTU Physics

PaNRAID School · supported by PEPR DIADEM



What we'll cover

Four techniques in focus, plus a map of everything else on offer.

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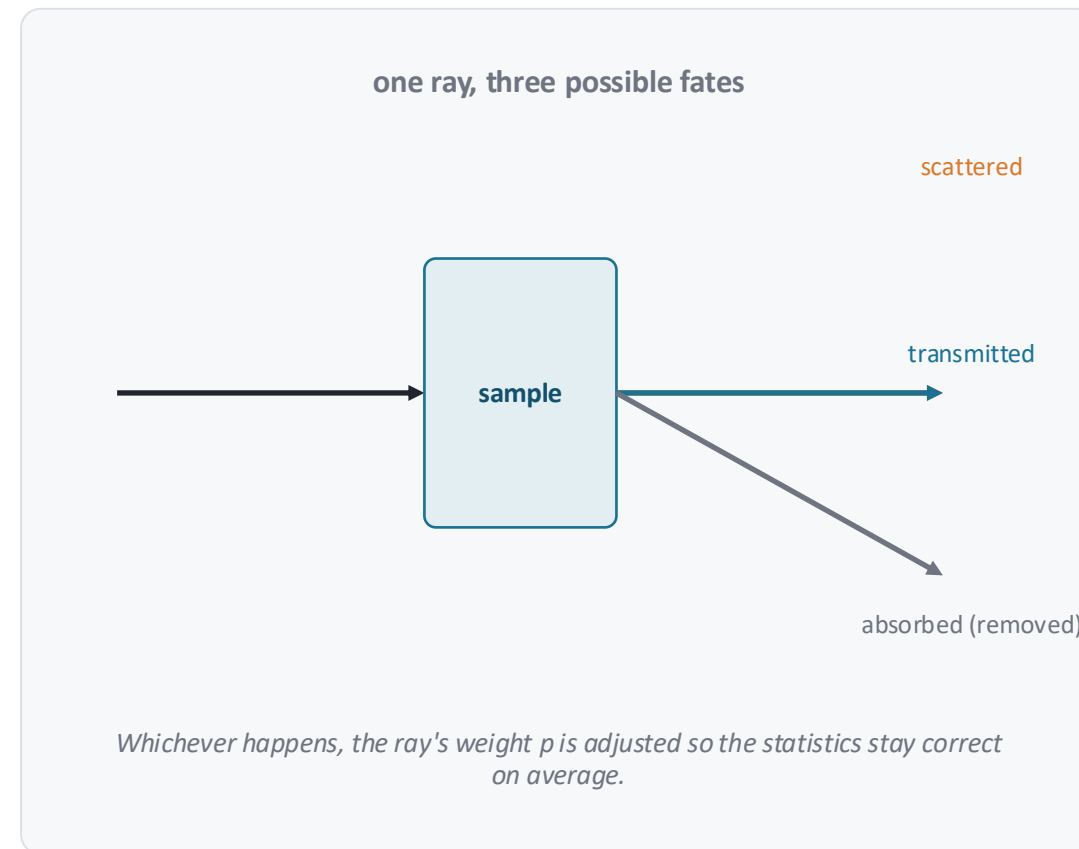
The full picture

One table, every technique, both codes



What a sample component does

- A sample intercepts the ray, decides — statistically — whether it scatters, transmits, or is absorbed, and updates the ray's state accordingly.
- `p_interact` lets you force a chosen fraction of rays to scatter, improving statistics without biasing the physical intensity.
- Most samples support several shapes out of the box: box, cylinder, sphere — or a hollow version of each via thickness.
- Many also accept an arbitrary OFF/PLY mesh (`geometry=file`) for a truly custom shape — the same complex-geometry engine underneath the diffraction, spectroscopy and imaging samples we'll see today.





The sample menu, at a glance

Four techniques get the spotlight today — with a map of what else exists at the end.

Diffraction

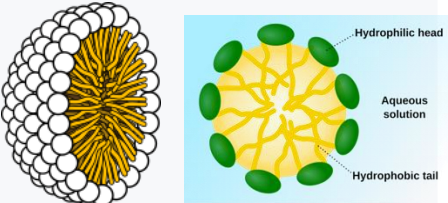
Powders & single crystals.
Both probes, both codes.



NEUTRON + X-RAY

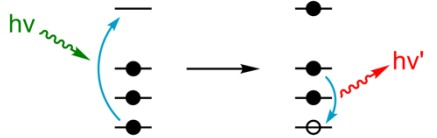
SANS / SAXS

Molecular-scale structure
from small-angle scattering.



NEUTRON + X-RAY

X-ray Spectroscopy

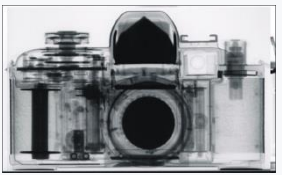


Absorption edges and
fluorescence lines.

X-RAY ONLY

Imaging

Absorption-contrast
radiography & tomography.



NEUTRONS

X-ray imaging can of course
also be done – but is left out
of PaNRAID to reduce school
scope...



X-RAYS
Left out of PaNRAID to
reduce school scope...

...and more

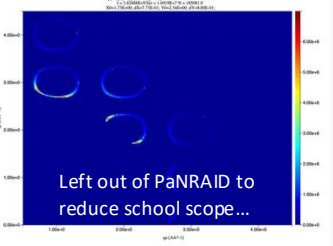
Reflectometry, inelastic
scattering, generic &
advanced tools.

All the rest!

SEE SUMMARY

Inelastic neutron
Scattering

Phonons, Magnons,
Spinwaves...



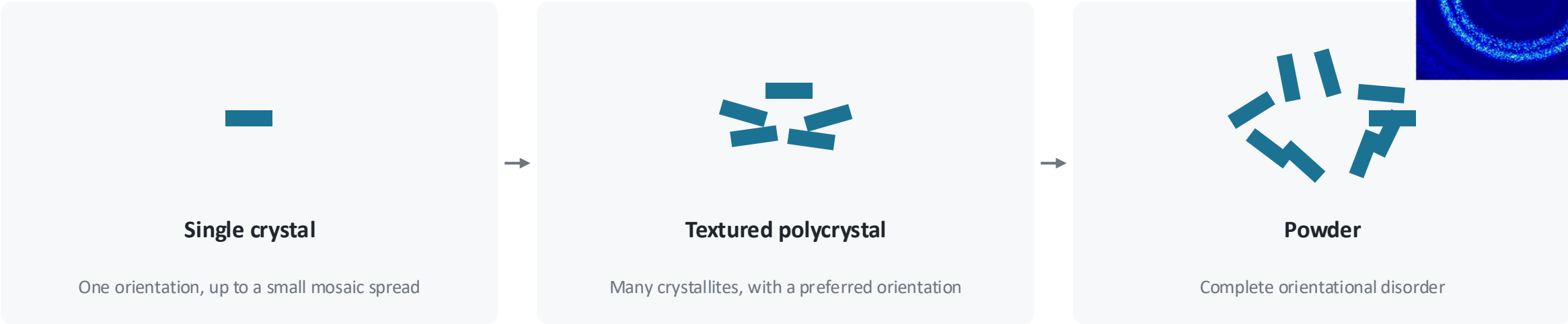
Left out of PaNRAID to
reduce school scope...

NEUTRON ONLY

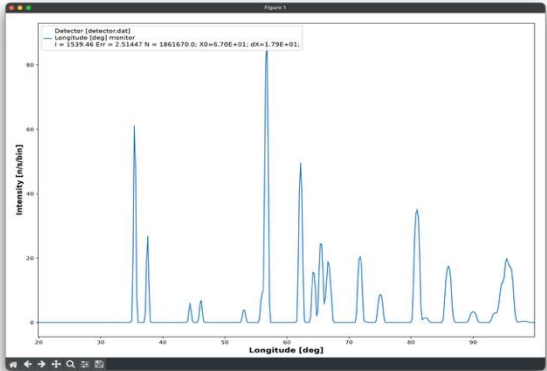
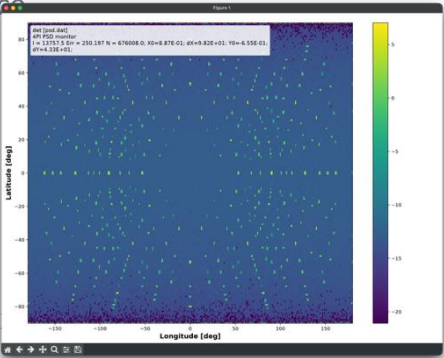


From single crystal to powder

A spectrum of disorder — the same underlying lattice, sampled differently.



McStas's Single_crystal component sits at the ordered end and can itself model a mosaic spread; PowderN sits at the fully disordered end — both exist, under the same names, in McXtrace.





Powder diffraction: PowderN

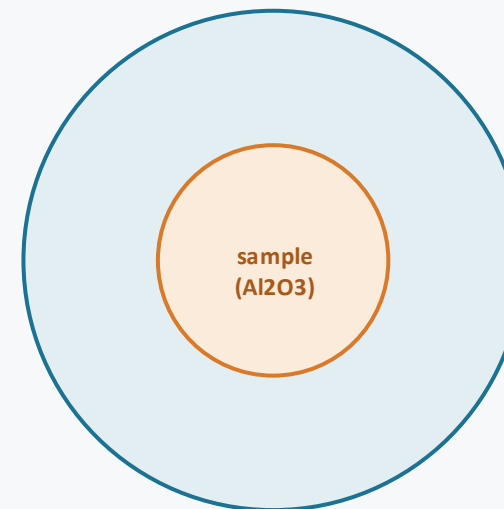
The same component, same reflection-file format, in both McStas and McXtrace.

- Reads a list of (hkl, d-spacing, structure factor) reflections from a data file — the physics is entirely data-driven.
- `concentric=1` lets a PowderN act as an outer container (e.g. a sample can) around an inner sample, sharing one coordinate position.
- `d_phi` and related focusing parameters restrict scattering to a chosen slice of 4π — useful when only a detector bank at certain angles exists.

```
COMPONENT can = PowderN(reflections="Al.laz",
    radius=0.02, yheight=0.05, thickness=0.005,
    concentric=1)
COMPONENT sample = PowderN(reflections="Al2O3.laz",
    ...)
COMPONENT can2 = COPY(can) (concentric=0)
```

a concentric container

can (Al)



Each layer computes its own scattering — the can's diffraction lines and the sample's appear together, correctly weighted.



Single-crystal diffraction

Single_crystal models an ideally imperfect crystal — peak broadening dominated by mosaic, not appropriate for near-perfect crystals.

Orientation: unit-cell vectors (ax,ay,az, bx,by,bz, cx,cy,cz) encode the crystal's orientation relative to the component's own frame — no assumption about cell shape or symmetry is made.

Isotropic mosaic

Gaussian spread of rotation angle around a random axis — one mosaic parameter for the whole crystal.

Anisotropic mosaic

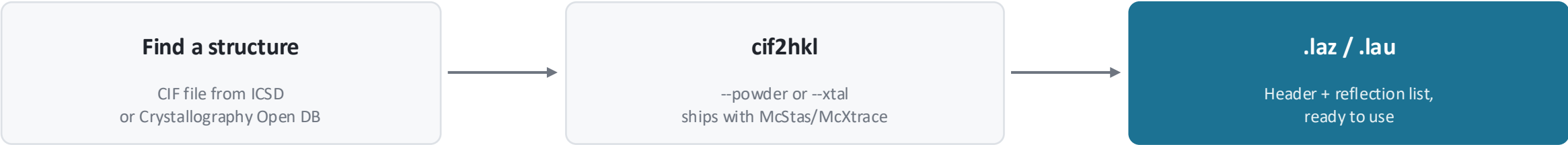
A separate Gaussian spread around each unit-cell axis (mosaic_a, mosaic_b, mosaic_c).

Rocking-curve mosaic

Mosaic derived from the measured widths of two Bragg peaks — give σ and hkl for each.



Getting reflection data



Used by: PowderN, Single_crystal, Isotropic_Sqw

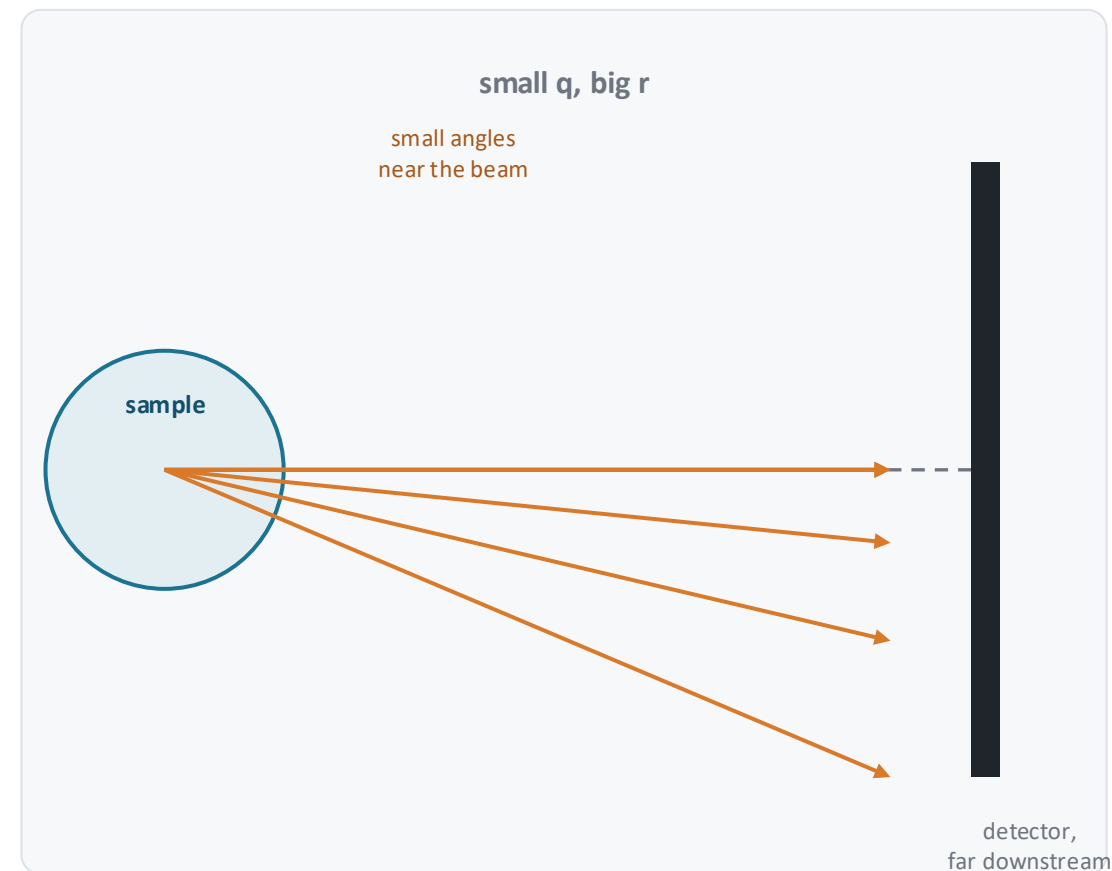
Data formats in McStas & McXtrace

Format	Used by
.lau	Single_crystal, PowderN, Isotropic_Sqw
.laz	PowderN, Isotropic_Sqw
.sqw / qSq	Isotropic_Sqw (qSq mostly for liquids)
PDB	SANSPDB / SAXSPDB
ncmat	NCrystal-backed samples (McStas)



Small-angle scattering, the shared idea

- Elastic scattering at small angles — small scattering vector q means sensitivity to big real-space distances r .
- Probes molecular-scale structure, typically 10–100 Å: proteins, polymers, nanoparticles, pores.
- Often a dilute molecule-in-solvent system: a small contrast in scattering density against a buffer/background.
- Usually isotropic scattering — the signal only depends on $|q|$, not its direction.
- Needs good collimation and a long flight path after the sample, to resolve the small angles involved.





SANS & SAXS models, side by side

Three tiers of model, available under matching names in both codes.

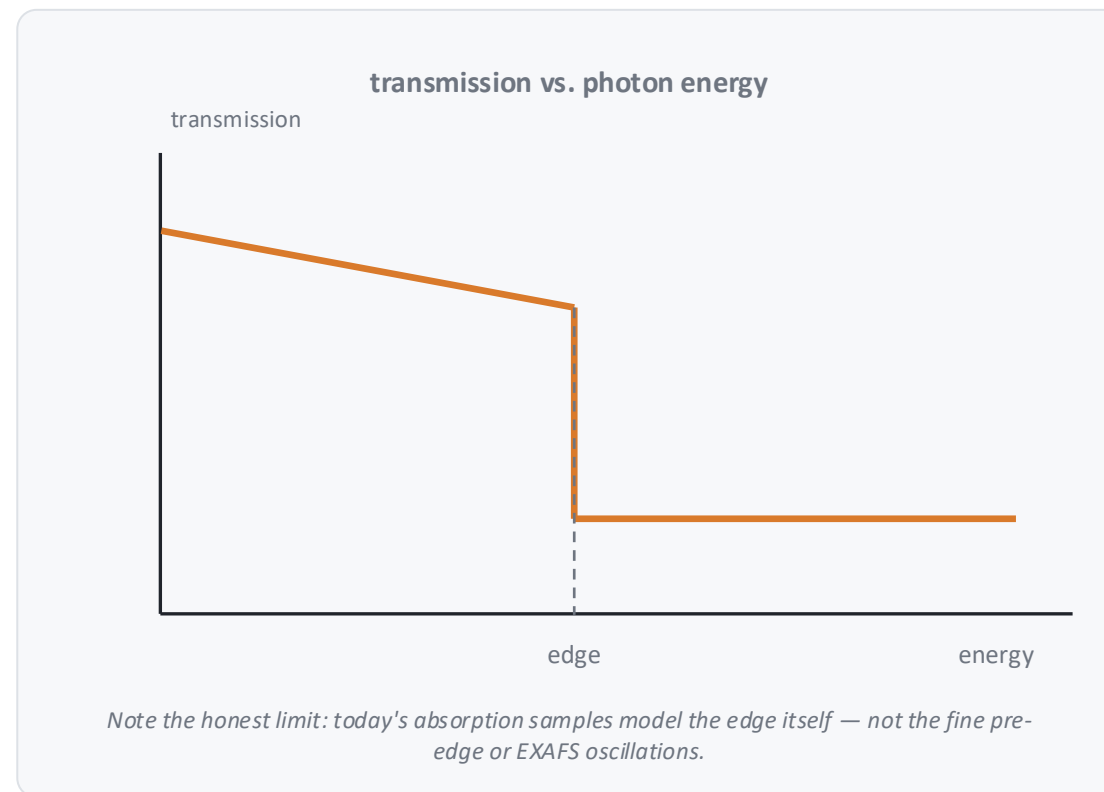
Tier	McStas (SANS)	McXtrace (SAXS)	Notes
Analytic shapes	<code>Sans_spheres</code>	<code>Saxs_spheres</code>	Exact naming parallel; dilute, monodisperse hard spheres, given contrast & radius
Benchmarked set	<code>SANS_benchmark2</code>	—	Experimentally-benchmarked model set (McStas, contrib library)
Full sasmodels library	<code>SasView_*</code> (92 components)	<code>SasView_*</code> (92 components)	File-for-file identical set in both codes — sphere, cylinder, core-shell, fractal, ...
Macromolecules	<code>SANSPDB</code>	<code>SAXSPDB</code>	Scattering directly from a PDB structure file (contrib library, both codes)

Under the hood: the sasmodels components are generated from the same SasView/sasmodels project (sasview.org) for both codes — 92 identically-named .comp files in mcstas-comps/sasmodels and mcxtrace-comps/sasmodels, sphere through triaxial ellipsoid.



Absorption & fluorescence: the physics

- A photon excites a core electron (e.g. the K shell). If the photon energy exceeds the binding energy, the electron is ejected — a photoelectron.
- The resulting core hole is filled by an electron from a higher shell — that transition emits a characteristic fluorescence line.
- At the absorption edge, the transmitted intensity drops sharply, and a fluorescence line appears at a lower, element-specific energy.
- Elastic (Rayleigh) and inelastic (Compton) scattering can accompany absorption — all part of the same interaction budget.





The Fluorescence component & friends

Fluorescence

- Models all four processes at once: absorption, fluorescence, Compton and Rayleigh scattering
- Any shape (box, cylinder, sphere, or OFF/PLY mesh) and any compound material, e.g. “CaCO3” or “ZnCu”
- Computed via XRayLib — about 10× slower than the simpler components, in exchange for real chemistry
- FluoCrystal and FluoPowder specialise the same physics for crystalline and powder samples

Filter, Abs_objects, Absorption_sample

- Absorption only — no fluorescence, Compton or Rayleigh
- Require pre-made absorption data files, and only handle pure, mono-atomic materials
- Abs_objects and Absorption_sample can combine several OFF/PLY shapes/materials in one component
- Much faster — the right choice when fluorescence itself isn't of interest

All of these support concentric geometry (a sample inside a can or cryostat) the same way PowderN does.



Two beamlines, built from these samples

ROCK · absorption edge

- Bending magnet source, 4–40 keV white beam
- A channel-cut DCM (two Bragg_crystal, Si 111) selects the energy
- A Fluorescence sample (“ZnCu”, 2×10×1 mm) sits after the monochromator
- An energy monitor after the sample shows the absorption shadow and the fluorescence lines together
- Scanning energy 8–10 keV sweeps straight across the Zn/Cu edges

LUCIA · hyperspectral fluorescence

- HU52 undulator source, feeding a 0.5×0.5 mm slit and a Si(111) DCM
- A Fluorescence sample stage — a wire-shaped inclusion (geometry="wire.ply") inside a different bulk material
- The two materials are combined so only one “fires” per ray — an advanced-grammar detail we won't dwell on here
- A position-resolved energy monitor turns the result into a hyperspectral map — composition read out from fluorescence lines, point by point



Neutron imaging

Absorption-contrast radiography and tomography — built from the same shapes as everything else.

- A single-phase absorbing “block” — box, cylinder, sphere, or an arbitrary OFF/PLY mesh — stands in for the imaged object.
- The same complex-geometry engine used for diffraction and spectroscopy samples renders detailed shapes: a chess piece, a bracket with drilled holes, and so on.
- Contrast comes purely from attenuation — how much of the beam a given path length and material absorbs.
- Rotate the object (or the source/detector) between exposures to build up a tomographic reconstruction.

On the roadmap: multi-phase materials, refractive effects, and phase-contrast imaging are being developed but aren't ready yet — today's imaging samples are single-phase, absorption-contrast only.

Why neutron-only in this session? X-ray imaging exists in principle (attenuation is attenuation) — but McXtrace's sample library so far has focused on spectroscopy and scattering rather than dedicated imaging phantoms.



Reflectometry & inelastic scattering

Reflectometry

- Probes surfaces & interfaces, solid or liquid, via specular and off-specular reflection
- McStas: Multilayer_Sample — phase-matrix formalism plus an incoherent background
- Or the simpler route: attach a measured reflectivity curve directly to a Mirror-type component
- Today's choice is deliberately small — not a rich library like SANS

Inelastic scattering $S(q, \omega)$

- McStas: Phonon_simple, Magnon_bcc, Isotropic_Sqw (inelastic mode), Single_crystal_inelastic, Tunneling_sample
- McXtrace: Isotropic_Sqw in an experimental Thomson (IXS) mode — no RIXS or single-crystal inelastic yet
- Computationally demanding: only a small fraction of rays land in a given (q, ω) window, so large ray counts are needed



Generic tools & advanced assemblies

Resolution samples

Res_sample, TOFRes_sample, Spot_sample — synthetic, delta-function-like scatterers used to study an instrument's resolution function, independent of any real material.

Union framework

Small physics-process components — Incoherent_process, Powder_process, Single_crystal_process, NCrystal_process (McStas); Incoherent_process, Powder_process, plus X-ray-specific Compton_xrl_process & Rayleigh_xrl_process (McXtrace) — combined via Union_master into complex multi-material assemblies. Present in both codes; its assembly grammar is a topic for a dedicated advanced session.

sasmodels & PDB

92 identically-named SasView_* components in both codes' sasmodels library, plus the SANSPDB / SAXSPDB macromolecule components — already introduced under SANS/SAXS, noted here for completeness.

A note on advanced grammar: concentric, GROUP, WHEN and EXTEND all let you combine sample components into richer assemblies. We've kept today's overview to the physics — ask if you'd like a pointer into the syntax.



The full picture

Technique	McStas	McXtrace	Notes
Powder diffraction	PowderN	PowderN	Shared name & file format
Single-crystal diffraction	Single_crystal	Single_crystal	Shared name; mosaic model
Isotropic structure/dynamics	Isotropic_Sqw	Isotropic_Sqw (exp.)	Elastic or inelastic, $S(q)$ / $S(q,\omega)$
Library-backed diffraction	NCrystal_sample	—	Via the NCrystal library
SANS / SAXS (analytic)	Sans_spheres	Saxs_spheres	Exact naming parallel
SANS / SAXS (full library)	SasView_* (92 comps)	SasView_* (92 comps)	File-for-file identical set
SANS / SAXS (macromolecule)	SANSPDB	SAXSPDB	Direct from a PDB file (contrib)
Absorption spectroscopy	—	Filter, Abs_objects, Absorption_sample	X-ray only; mono-atomic, fast
Fluorescence	—	Fluorescence, FluoCrystal, FluoPowder	X-ray only; any shape/compound
Imaging	absorbing blocks / OFF meshes	—	Neutron only, in practice
Reflectometry	Multilayer_Sample	—	Small library (contrib)
Inelastic scattering	Phonon_simple, Magnon_bcc, ...	Isotropic_Sqw (IXS, exp.)	Richer on the neutron side
Resolution / generic	Res_sample, TOFRes_sample, Spot_sample	—	Synthetic, for resolution studies
Advanced assemblies	Union components	Union components	Shared framework, both codes

Where to go deeper

McStas

`mcdoc sample / mcdoc -c`

Data: `$MCSTAS/data` (Lau/Laz/Sqw/PDB/ncmat)

`cif2hkl --powder / --xtal file.cif`

McXtrace

`mxdoc sample / mxdoc -c`

Practicals: absorption (ROCK), fluorescence (LUCIA)

McCode wiki: generating Single_crystal/PowderN files

Peter Willendrup · pkwi@fysik.dtu.dk · DTU Physics